Statistical APT Model using PCA

Extracting Factors from Returns with the SVD

IEOR E4576 — Data-Driven Models in Finance — Fall 2026 — Week 8

1 The APT Model

On each day $t = 1, \ldots, T$ the returns of n assets satisfy

$$\mathbf{R}_t = \boldsymbol{\mu} + \mathbf{B}\mathbf{f}_t + \boldsymbol{\epsilon}_t \tag{1}$$

where $\mathbf{R}_t \in \mathbb{R}^{n\times 1}$ is the vector of returns on day t, $\boldsymbol{\mu} \in \mathbb{R}^{n\times 1}$ is the vector of expected returns, $\mathbf{B} \in \mathbb{R}^{n\times k}$ is the matrix of factor loadings (row i holds the loadings of asset i), $\mathbf{f}_t \in \mathbb{R}^{k\times 1}$ is the vector of factor realizations on day t, and $\boldsymbol{\epsilon}_t \in \mathbb{R}^{n\times 1}$ is the vector of idiosyncratic errors. The factors and errors have mean zero and are uncorrelated with each other, with

$$\mathrm{Cov}(\mathbf{f}_t) = \boldsymbol{\Sigma}_F \in \mathbb{R}^{k\times k}$$

$$\mathrm{Cov}(\boldsymbol{\epsilon}_t) = \mathbf{D} \in \mathbb{R}^{n\times n} \text{ diagonal}$$

Only the returns $\mathbf{R}_t$ are observed. Neither the loadings $\mathbf{B}$ nor the factors $\mathbf{f}_t$ are observed; both are estimated from the returns.

2 Covariance Structure

Since $\boldsymbol{\mu}$ is constant and $\mathbf{f}_t$ and $\boldsymbol{\epsilon}_t$ are uncorrelated, (1) implies

$$\boldsymbol{\Sigma} = \mathrm{Cov}(\mathbf{R}_t) = \mathrm{Cov}(\mathbf{B}\mathbf{f}_t) + \mathrm{Cov}(\boldsymbol{\epsilon}_t)$$

that is,

$$\boldsymbol{\Sigma} = \mathbf{B}\boldsymbol{\Sigma}_F\mathbf{B}^T + \mathbf{D} \tag{2}$$

The covariance matrix of returns is a rank-k matrix plus a diagonal matrix.

3 Estimation with the SVD

Collect the returns of the T days as the columns of

$$\mathbf{R} = \begin{bmatrix}\mathbf{R}_1 & \cdots & \mathbf{R}_T\end{bmatrix} \in \mathbb{R}^{n\times T}$$

Estimate $\boldsymbol{\mu}$ by the sample mean return of each asset, $\hat{\boldsymbol{\mu}} = \frac{1}{T}\mathbf{R}\mathbf{1}_T$, where $\mathbf{1}_T \in \mathbb{R}^{T\times 1}$ is the vector of ones, and subtract it from each column:

$$\tilde{\mathbf{R}} = \mathbf{R} - \hat{\boldsymbol{\mu}}\mathbf{1}_T^T$$

Substituting (1) for each column, $\boldsymbol{\mu}$ cancels and

$$\tilde{\mathbf{R}} = \mathbf{B}\mathbf{F} + \mathbf{E} \tag{3}$$

where

$$\mathbf{F} = \begin{bmatrix}\mathbf{f}_1 - \bar{\mathbf{f}} & \cdots & \mathbf{f}_T - \bar{\mathbf{f}}\end{bmatrix} \in \mathbb{R}^{k\times T}$$

$$\mathbf{E} = \begin{bmatrix}\boldsymbol{\epsilon}_1 - \bar{\boldsymbol{\epsilon}} & \cdots & \boldsymbol{\epsilon}_T - \bar{\boldsymbol{\epsilon}}\end{bmatrix} \in \mathbb{R}^{n\times T}$$

and $\bar{\mathbf{f}}$ and $\bar{\boldsymbol{\epsilon}}$ are the sample means of the factors and errors.

Take the SVD of the de-meaned returns,

$$\tilde{\mathbf{R}} = \mathbf{U}\mathbf{S}\mathbf{V}^T \tag{4}$$

with $\mathbf{U} \in \mathbb{R}^{n\times n}$ and $\mathbf{V} \in \mathbb{R}^{T\times T}$ orthogonal and $\mathbf{S} \in \mathbb{R}^{n\times T}$ diagonal with entries $s_1 \geq s_2 \geq \cdots \geq 0$. Let $\mathbf{U}_k \in \mathbb{R}^{n\times k}$ and $\mathbf{V}_k \in \mathbb{R}^{T\times k}$ be the first k columns of $\mathbf{U}$ and $\mathbf{V}$, and $\mathbf{S}_k = \mathrm{diag}(s_1, \ldots, s_k)$. Keeping the first k terms gives the best rank-k approximation of $\tilde{\mathbf{R}}$:

$$\tilde{\mathbf{R}} = \mathbf{U}_k\mathbf{S}_k\mathbf{V}_k^T + \hat{\mathbf{E}}$$

Matching this with (3) gives the estimates

$$\hat{\mathbf{B}} = \mathbf{U}_k \in \mathbb{R}^{n \times k} \tag{5}$$

$$\hat{\mathbf{F}} = \mathbf{S}_k \mathbf{V}_k^T \in \mathbb{R}^{k \times T} \tag{6}$$

with residuals $\hat{\mathbf{E}} = \tilde{\mathbf{R}} - \hat{\mathbf{B}}\hat{\mathbf{F}}$. The estimated factor covariance and idiosyncratic variances are

$$\hat{\mathbf{\Sigma}}_F = \frac{1}{T-1}\hat{\mathbf{F}}\hat{\mathbf{F}}^T = \frac{1}{T-1}\mathbf{S}_k^2$$

$$\hat{\mathbf{D}} = \operatorname{diag}\left(\frac{1}{T-1}\hat{\mathbf{E}}\hat{\mathbf{E}}^T\right)$$

and substituting into (2) gives the estimate of the covariance matrix

$$\hat{\mathbf{\Sigma}}_{PCA} = \hat{\mathbf{B}}\hat{\mathbf{\Sigma}}_F\hat{\mathbf{B}}^T + \hat{\mathbf{D}} = \frac{1}{T-1}\left(\mathbf{U}_k\mathbf{S}_k^2\mathbf{U}_k^T + \operatorname{diag}\left(\hat{\mathbf{E}}\hat{\mathbf{E}}^T\right)\right) \tag{7}$$

4 Relation to the Eigendecomposition of the Covariance Matrix

The sample covariance matrix of returns is

$$\hat{\mathbf{\Sigma}} = \frac{1}{T-1}\tilde{\mathbf{R}}\tilde{\mathbf{R}}^T$$

Substituting the SVD (4) and using $\mathbf{V}^T\mathbf{V} = \mathbf{I}$:

$$\hat{\mathbf{\Sigma}} = \frac{1}{T-1}\mathbf{U}\mathbf{S}\mathbf{S}^T\mathbf{U}^T = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T \tag{8}$$

where $\mathbf{\Lambda} = \frac{1}{T-1}\mathbf{S}\mathbf{S}^T$ is diagonal with entries $\lambda_j = s_j^2/(T-1)$. This is the eigendecomposition of $\hat{\mathbf{\Sigma}}$. Therefore:

- The loadings $\hat{\mathbf{B}} = \mathbf{U}_k$ are the eigenvectors of $\hat{\mathbf{\Sigma}}$ for its k largest eigenvalues.
- The factor covariance is $\hat{\mathbf{\Sigma}}_F = \mathbf{\Lambda}_k = \operatorname{diag}(\lambda_1, \ldots, \lambda_k)$, the k largest eigenvalues.
- The factors are the projections of the returns on the eigenvectors, the principal components: $\hat{\mathbf{F}} = \mathbf{U}_k^T\tilde{\mathbf{R}} = \mathbf{S}_k\mathbf{V}_k^T$.

The estimate (7) keeps the top k terms of the eigendecomposition (8) and replaces the remaining $n-k$ terms by the diagonal matrix $\hat{\mathbf{D}}$:

$$\hat{\mathbf{\Sigma}}_{PCA} = \mathbf{U}_k\mathbf{\Lambda}_k\mathbf{U}_k^T + \hat{\mathbf{D}}$$

5 What the Truncation Drops

The SVD writes $\tilde{\mathbf{R}}$ exactly as $\mathbf{USV}^T$. The estimate keeps the first k columns of $\mathbf{U}$, the top-left $k \times k$ block of $\mathbf{S}$ and the first k rows of $\mathbf{V}^T$ (blue), and drops the rest (gray). The kept blocks multiply to $\hat{\mathbf{B}}\hat{\mathbf{F}} = \mathbf{U}_k\mathbf{S}_k\mathbf{V}_k^T$; the dropped blocks multiply to the residual $\hat{\mathbf{E}}$.

